# PARISHRAM 2.0

## CLASS 12TH 2026

## Electrostatic Potential and Capacitance

PHYSICS | LECTURE-04

BY-DEV AGNANI SIR



# Topics to be covered:

1. Capacitors and Capacitance
3. Combination of Capacitors in Series
3. Combination of Capacitors in Parallel

# RECAP:

1. Potential Energy in an External Field of a:
   1. Single Charge $U = qV$
   2. System of Two Charges
   3. Dipole
2. Conductors and Insulators
3. Free Charges and Bound Charges inside a Conductor
4. Dielectrics and Electric Polarization

Literally **Parishram 2.0** students to **Dev Sir** after **Mandar Sir's** joke about **Dev sir** getting engaged:

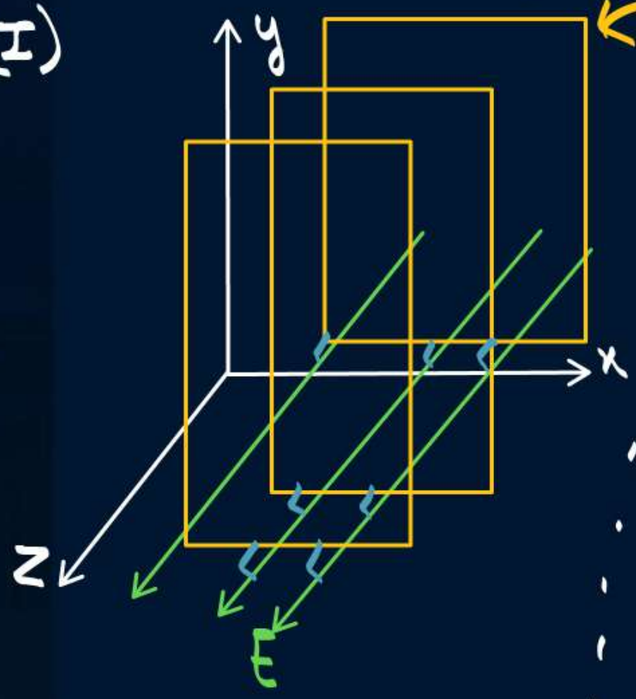


# QUESTION

1. Draw the equipotential surfaces corresponding to a uniform electric field in the z-direction. **(1 m)**
2. Draw an expression for the electric potential at any point along the axial line of an electric dipole. **(1 m)**

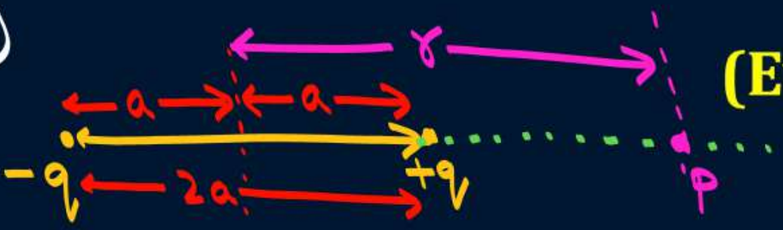
**(Ev (CBSE, 2019)**

(I)




Equipotential surfaces in the x-y plane.

(II)




$$V_p = \frac{kq}{(r-a)} + \frac{k(-q)}{(r+a)}$$

$$\Rightarrow V_p = kq\left[\frac{1}{(r-a)} - \frac{1}{(r+a)}\right] = kq\left[\frac{r+a-r+a}{r^2-a^2}\right]$$



$$V_p = \frac{Kq(2a)}{r^2 - a^2}$$

$$\boxed{V_p = \frac{Kp}{r^2 - a^2}}$$

$$W = -\int_{\theta_0}^{\theta_1} pE\sin\theta \cdot d\theta$$

$$W = -pE\left[-\cos\theta\right]_{\theta_0}^{\theta_1}$$

$$W = -pE\left[-\cos\theta_1 - (-\cos\theta_0)\right]$$

$$\boxed{W = pE\left[\cos\theta_1 - \cos\theta_0\right]}$$

$\theta_1$ : Final Angle

$\theta_0$ : Initial angle.

# Capacitors and Capacitance



→ Capacitance of a capacitor is a constant value.

$$C = \frac{Q}{V}$$

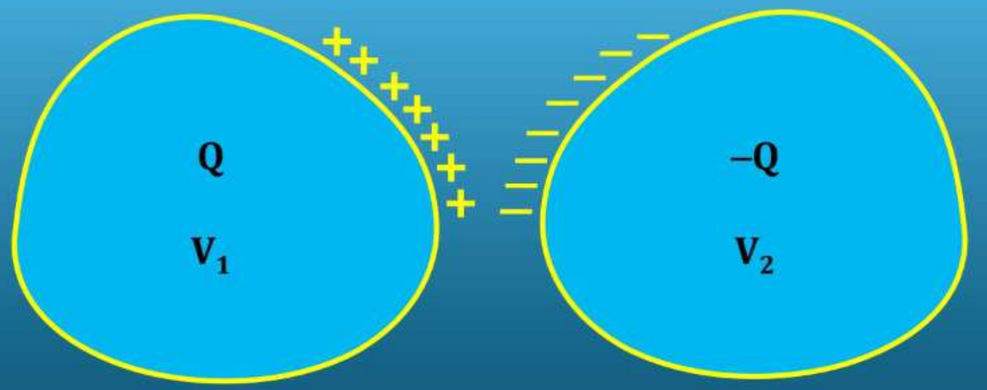
- Charge deposited on a capacitor is constant.
- Potential across a capacitor is also constant.

$$V = \frac{KQ}{r} \Rightarrow \frac{Q}{V} = \frac{r}{K}$$

$r \rightarrow$ Constant

$K \rightarrow$ Constant

$\Rightarrow$ Capacitance of a capacitor.




Conductor 1 Conductor 2

**A system of two conductors separated by an insulator forms a capacity.**

## ✡ S.I unit :

→ **Farad (F)**

→ $1\,F = \frac{1\,\text{Coulomb}}{1\,\text{Volt}}$

→ $F = CV^{-1}$

(*) $\mu F = 1 \times 10^{-6}\,F$

(*) $nF = 1 \times 10^{-9}\,F$

(*) $pF = 1 \times 10^{-12}\,F$

## Symbol of Capacitor :



} Constant Capacitor



} Variable Capacitor.

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$C = \frac{Q}{V}$ → $[AT]$

$V$ → $\frac{F}{q} \times d = \frac{[MLT^{-2}][L]}{[AT]}$

$\{i = \frac{q}{t}\}$

} $[M^{-1}L^{-2}T^{4}A^{2}]$ ← Dimensional Formula of Capacitance.

Current → $[A]$
Time → $[T]$
Mass → $[M]$
Length → $[L]$.

# Capacitors and Capacitance

1. A capacitor is a **two-terminal electric component**.
2. It has ability to **store energy in the form of electric charge**.
3. Capacitors are also called **Electric Condensers**.
4. They are designed to **enhance** and **increase capacitance**.
5. The **size** and **shape** of a **capacitor** affect its capacitance.



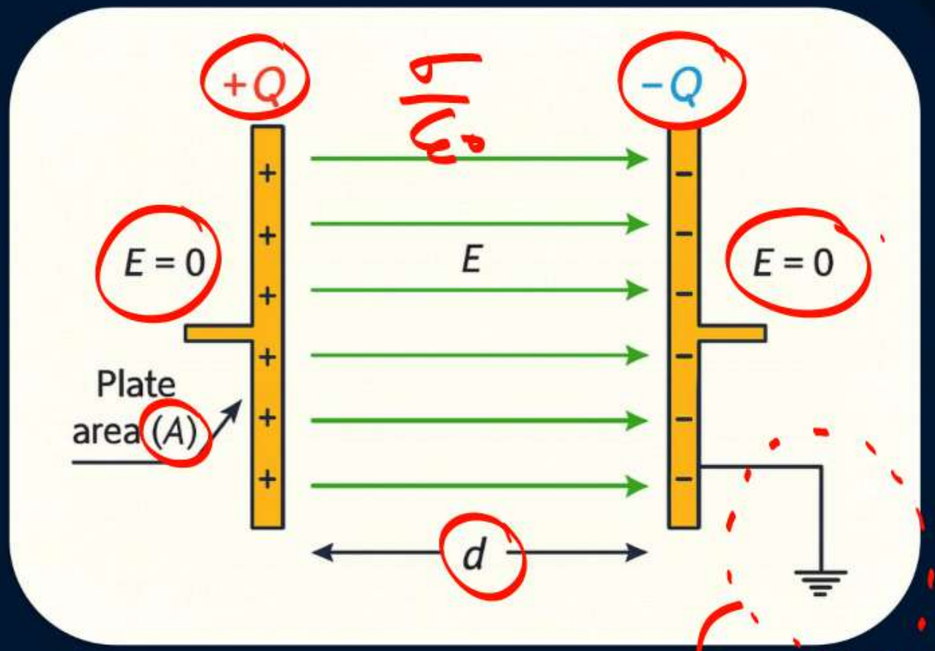
# Parallel Plate Capacitor

$$\left\{E = \frac{\sigma}{\epsilon_0}\right\} \qquad C = \frac{Q}{V} = \frac{Q}{Ed}$$

$$\Rightarrow C = \frac{Q\epsilon_0}{\sigma d}$$

$\sigma = \frac{Q}{A}$ — Surface charge density

$$\Rightarrow C = \frac{\cancel{Q}\epsilon_0 A}{\cancel{Q} d} \Rightarrow \boxed{C = \frac{\epsilon_0 A}{d}}$$




Earthed. (V=0)

# QUESTION

The capacity of parallel plate condenser depends on:

A the type of metal used

B the thickness of plates

C the potential difference applied across the plates

D the separation between the plates.



# QUESTION

2 marks

A parallel plate capacitor has rectangular plates of $400\ cm^2$ area and are separated by a distance of 2 mm. What charge will appear on the plates if a 200 volt potential difference is applied across the capacitor?

A $3.54 \times 10^{-6}\ C$

B $3.54 \times 10^{-8}\ C$

C $3.54 \times 10^{-10}\ C$

D $1770.8 \times 10^{-13}\ C$

$\frac{Q}{V} = \frac{\varepsilon_0 A}{d}$

$\varepsilon_0 = 8.85 \times 10^{-12}$

$Q = \frac{\varepsilon_0 A V}{d}$

$= \frac{8.85 \times 10^{-12} \times 400 \times 10^{-4} \times 200}{2 \times 10^{-3}}$

$= 8.85 \times 4 \times 10^{4} \times 10^{-13}$

$= 35. \times 10^{-9}$

$= 3.5 \times 10^{-8}$



# QUESTION

A 500 μF capacitor is charged at a rate of 125 μC/sec. The potential difference across the capacitor will be 10 V after an interval of,

Total charge = (125 μC/sec) × t

$C = \frac{Q}{V} \Rightarrow 500 \times 10^{-6} = \frac{125 \times 10^{-6} \times t}{10}$

$\Rightarrow t = 40\ s$

A 80 sec

B 50 sec

C 20 sec

D 40 sec



# QUESTION

If the distance between parallel plates of a capacitor is halved and dielectric constant is doubled then the capacitance

A Decreases two times

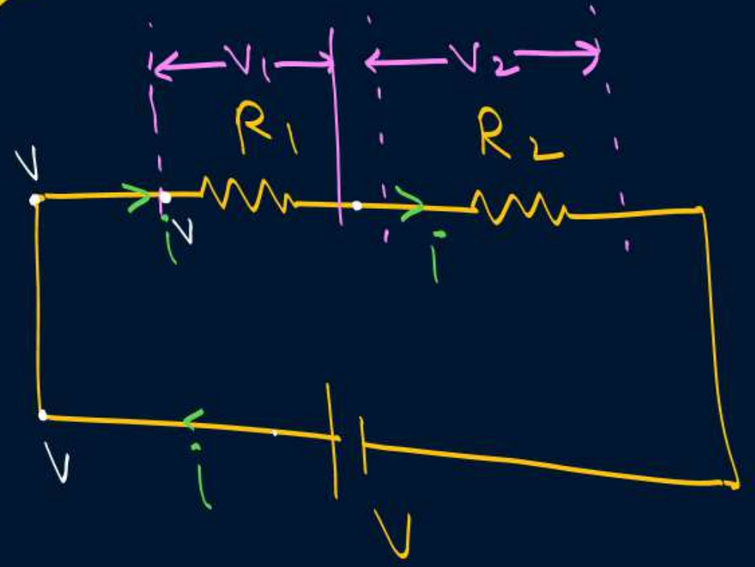
B Increases two times

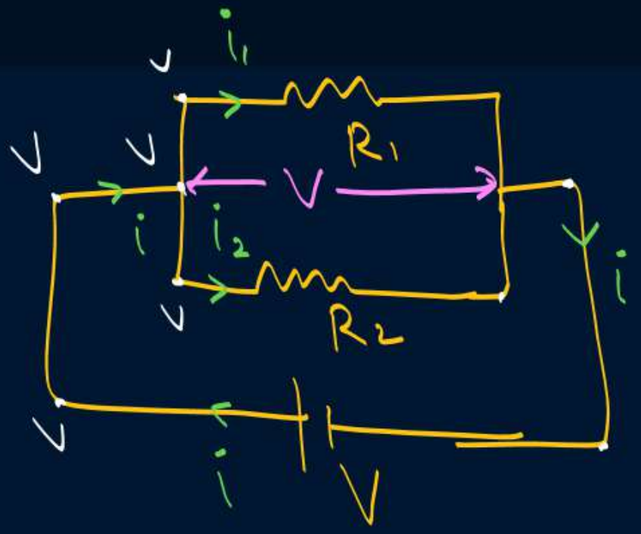
C Increases four times

D Remains the same.



# Class-10th Revision




Potential across both the resistors will be same and equal to the potential of the circuit.

$V = iR$

# Combination of Capacitors in Series

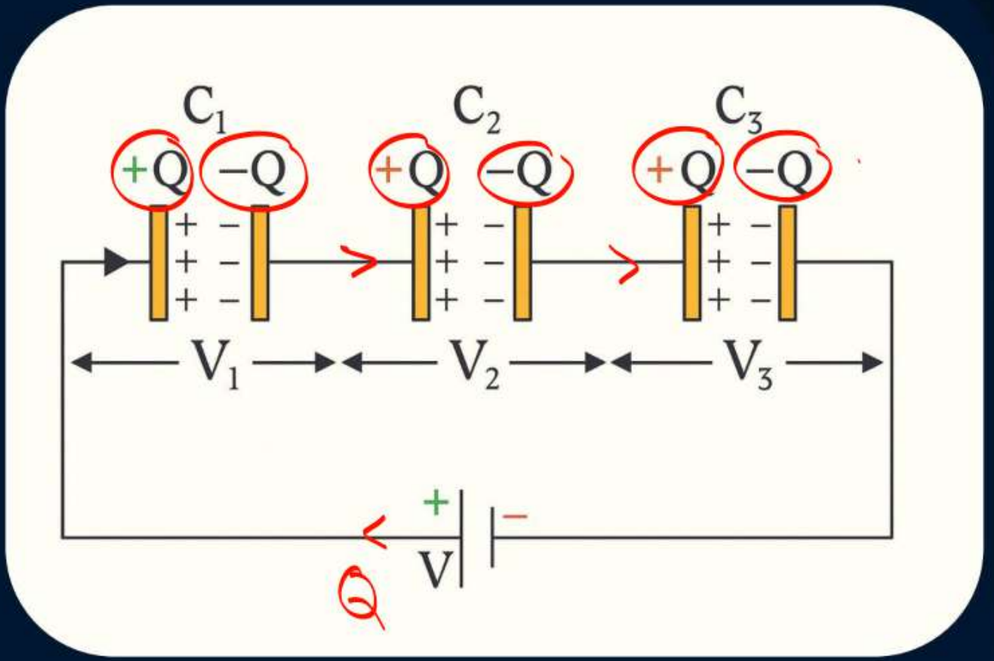
$$V = V_1 + V_2 + V_3$$

$$\Rightarrow \frac{Q}{C_{eq}} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$$

$$\left[ C = \frac{Q}{V} \Rightarrow V = \frac{Q}{C} \right]$$

$$\Rightarrow \boxed{\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$$

↑ Equivalent capacitance of the combination when capacitors $C_1, C_2$ and $C_3$ are connected in series.



# Combination of Capacitors in Series

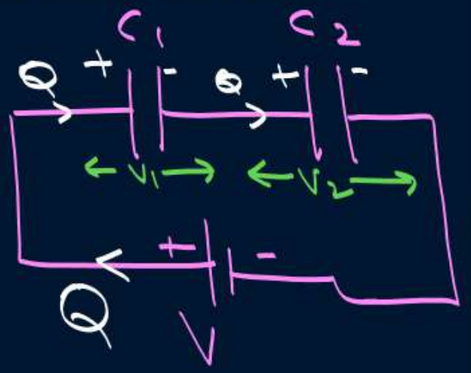
- If two capacitors having capacitances $C_1$ and $C_2$ are connected in series then :

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{C_1 + C_2}{C_1 C_2}$$

$$\Rightarrow C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

$$V_1 = \left(\frac{C_2}{C_1 + C_2}\right).V \text{ and } V_2 = \left(\frac{C_1}{C_1 + C_2}\right).V$$



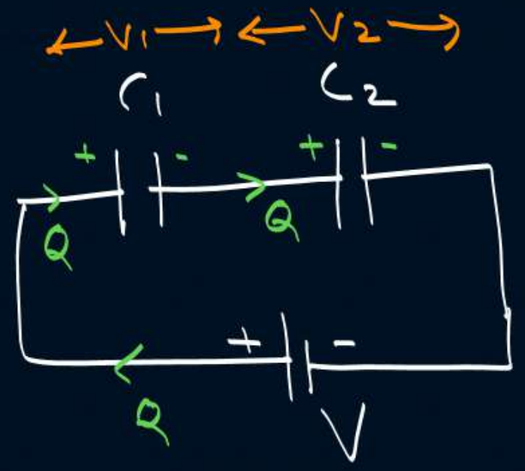

- If n identical capacitors each having capacitances C are connected in series with supply voltage V then Equivalent capacitance $C_{eq} = \frac{C}{n}$ and potential difference across each capacitor $V' = \frac{V}{n}$.

$(Q)_{C_1} = Q_{net}$

$\Rightarrow C_1 V_1 = C_{eq} \cdot V$

$\Rightarrow C_1 V_1 = \dfrac{C_1 C_2}{C_1 + C_2} \cdot V$

$\Rightarrow$ $\boxed{V_1 = \left(\dfrac{C_2}{C_1 + C_2}\right) \cdot V}$




$C = \dfrac{Q}{V}$

$V = V_1 + V_2$

$\Rightarrow V = \dfrac{C_2 V}{C_1 + C_2} + V_2$

$\Rightarrow$ $\boxed{V_2 = \dfrac{C_1 V}{C_1 + C_2}}$

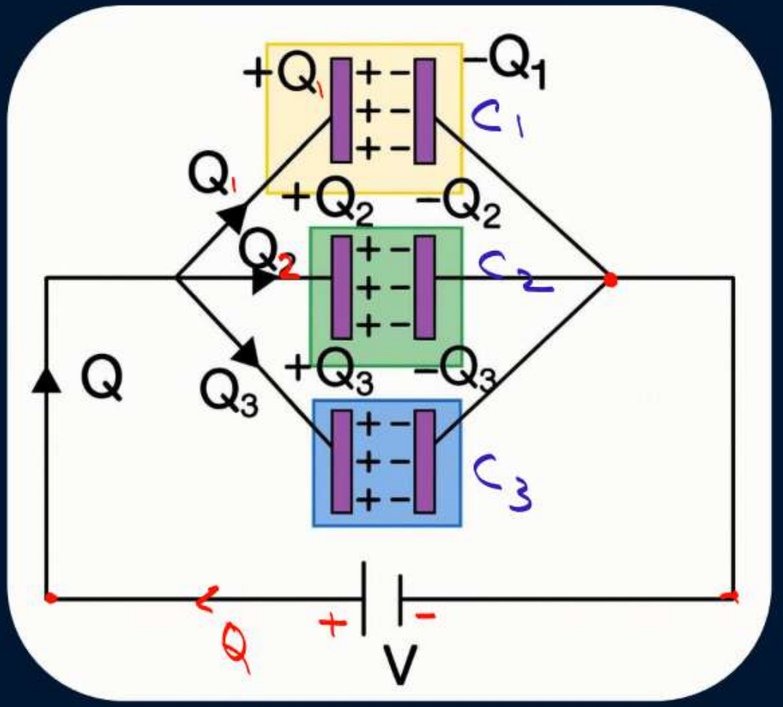
# Combination of Capacitors in Parallel

$$Q = Q_1 + Q_2 + Q_3$$

$$\Rightarrow (C_{eq.})(V) = (C_1)(V) + (C_2)(V) + (C_3)V$$

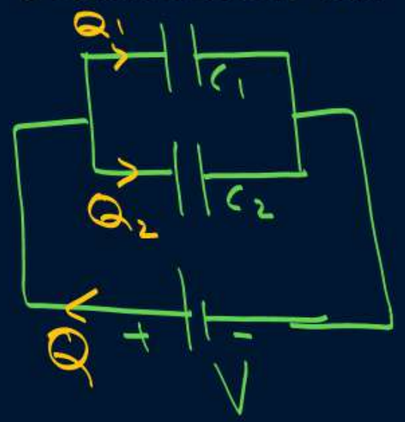
$$\Rightarrow C_{eq.} \cancel{V} = (C_1 + C_2 + C_3)\cancel{V}$$

$$\Rightarrow \boxed{C_{eq.} = C_1 + C_2 + C_3}$$



# Combination of Capacitors in Parallel

- If two capacitors having capacitances $C_1$ and $C_2$ respectively are connected in parallel then $C_{eq} = C_1 + C_2$.




$$Q_1 = \left(\frac{C_1}{C_1 + C_2}\right).Q \text{ and } Q_2 = \left(\frac{C_2}{C_1 + C_2}\right).Q$$

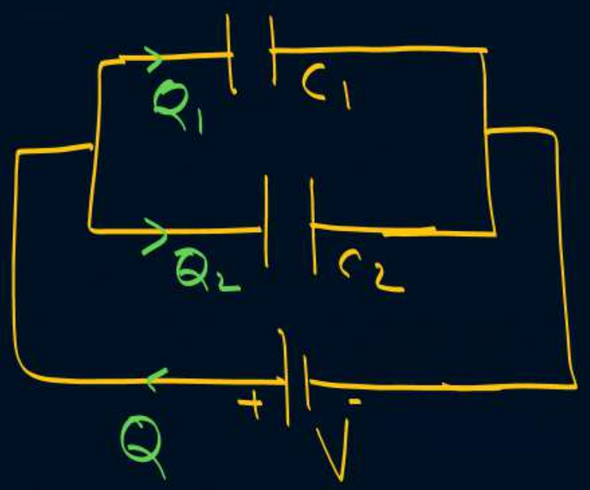
- If n identical capacitors are connected in in parallel, then Equivalent capacitance $C_{eq} = nC$ and charge on each $Q' = \frac{Q}{n}$.



$V_1 = V$

$\Rightarrow \frac{Q_1}{C_1} = \frac{Q}{C_{eq.}}$

$\Rightarrow \boxed{Q_1 = \frac{C_1}{C_1 + C_2} \cdot Q}$




$Q = Q_1 + Q_2$

$Q_2 = Q - Q_1$

$\Rightarrow \boxed{Q_2 = \frac{C_2}{C_1 + C_2} \cdot Q}$

# QUESTION

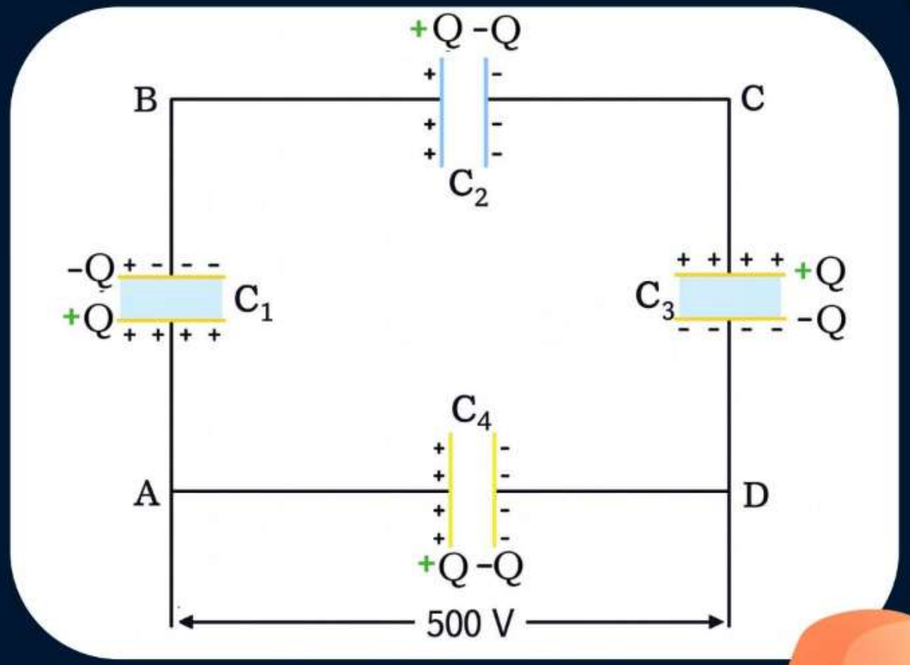
H.W

A network of four 10 μF capacitors is connected to a 500 V supply, as shown in figure. Determine:

(a) the equivalent capacitance of the network and

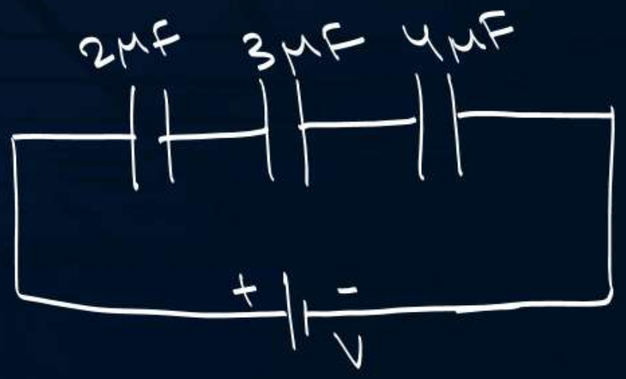
(b) the charge on each capacitor.

(**Note**, the charge on a capacitor is the charge on the plate with higher potential, equal and opposite to the charge on the plate with lower potential.)






Que:




Find $C_{eq.}$ ?

A) $\frac{3}{4}$

B) $\frac{18}{19}$

C) $\frac{12}{13}$

D) $\frac{14}{15}$

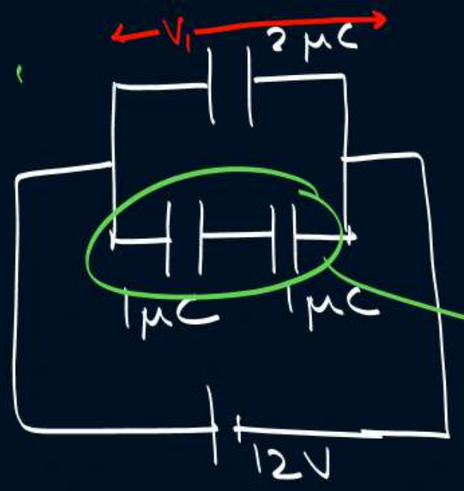
$$\frac{1}{C_{eq.}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C_{eq.}} = \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$$

$$= \frac{13}{12}$$

$C_{eq.} = \frac{12}{13}$

Que:




Find $C_{eq.}$, $V_1$ ?

$$\frac{1}{C_{eq.}} = \frac{1}{1} + \frac{1}{1} = 2$$

$$\Rightarrow C_{eq.} = \frac{1}{2}\mu F$$

$= 0.5\mu F$

$C_{eq.} = 2\mu F + 0.5\mu F$

$C_{eq.} = 2.5\mu F$

A) ✗ 2μF, 10V

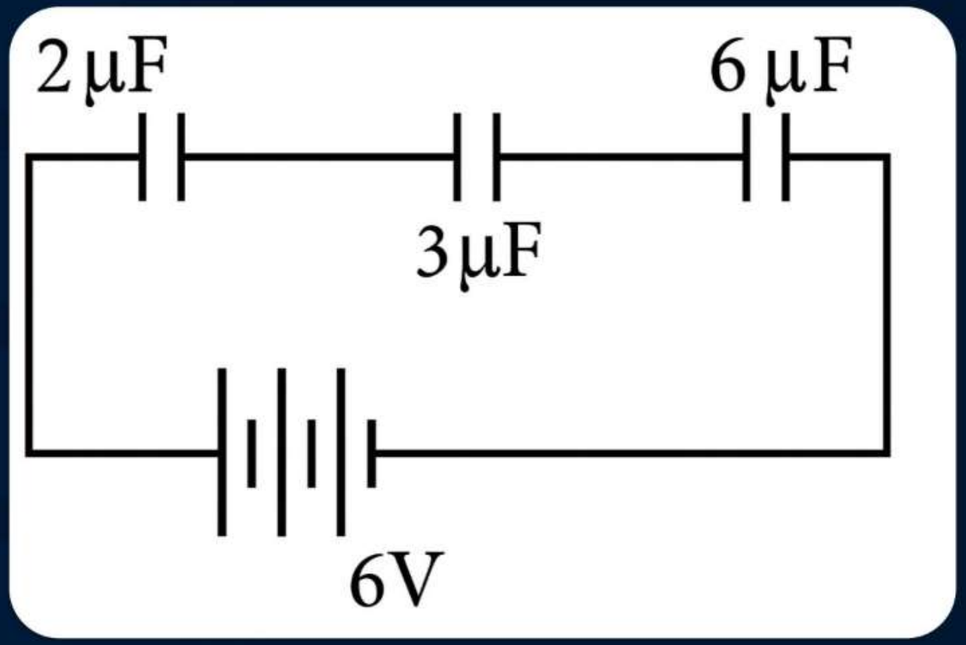
B) ✗ 2μF, 12V

C) ✗ 2.5μF, 10V

D) 2.5μF, 12V

# QUESTION

In the combination shown in the figure, the potential different across 3 μF will be:




A. 4.5 V

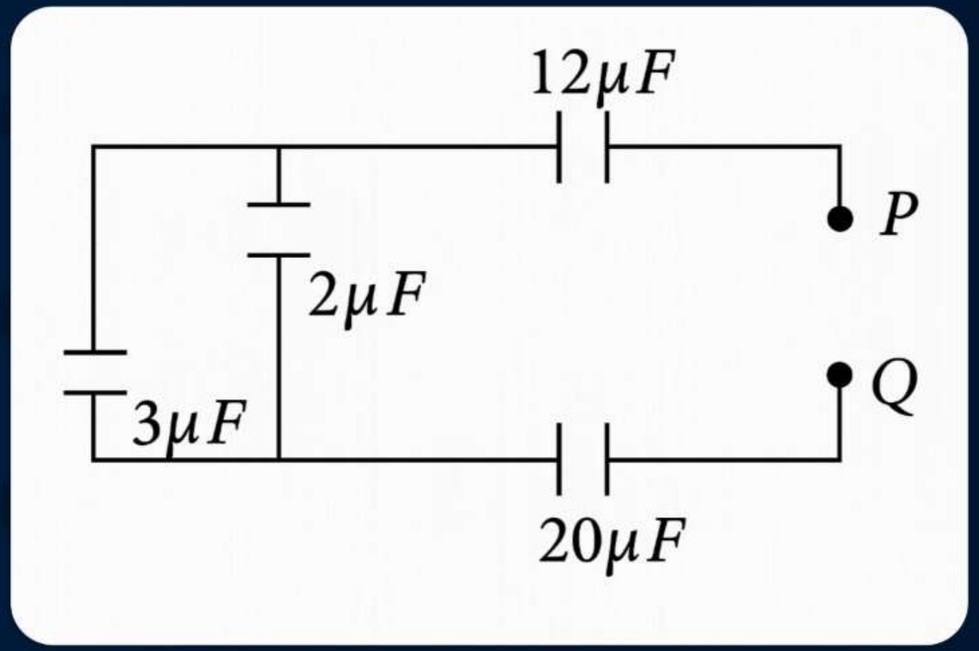
B. $\left(\frac{18}{11}\right)$ V

C. 3V

D. 2V

# QUESTION

In the circuit diagram shown in the adjoining figure, the resultant capacitance between P and Q is:




A 47μF

B 3 μF

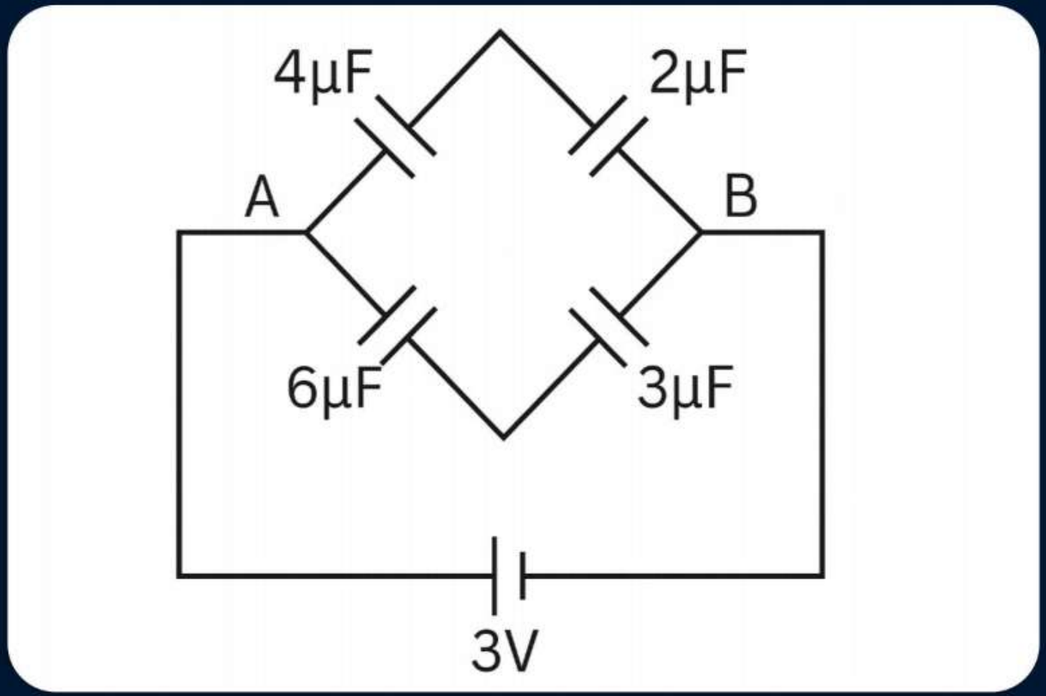
C 60 μF

D 10 μF

# QUESTION

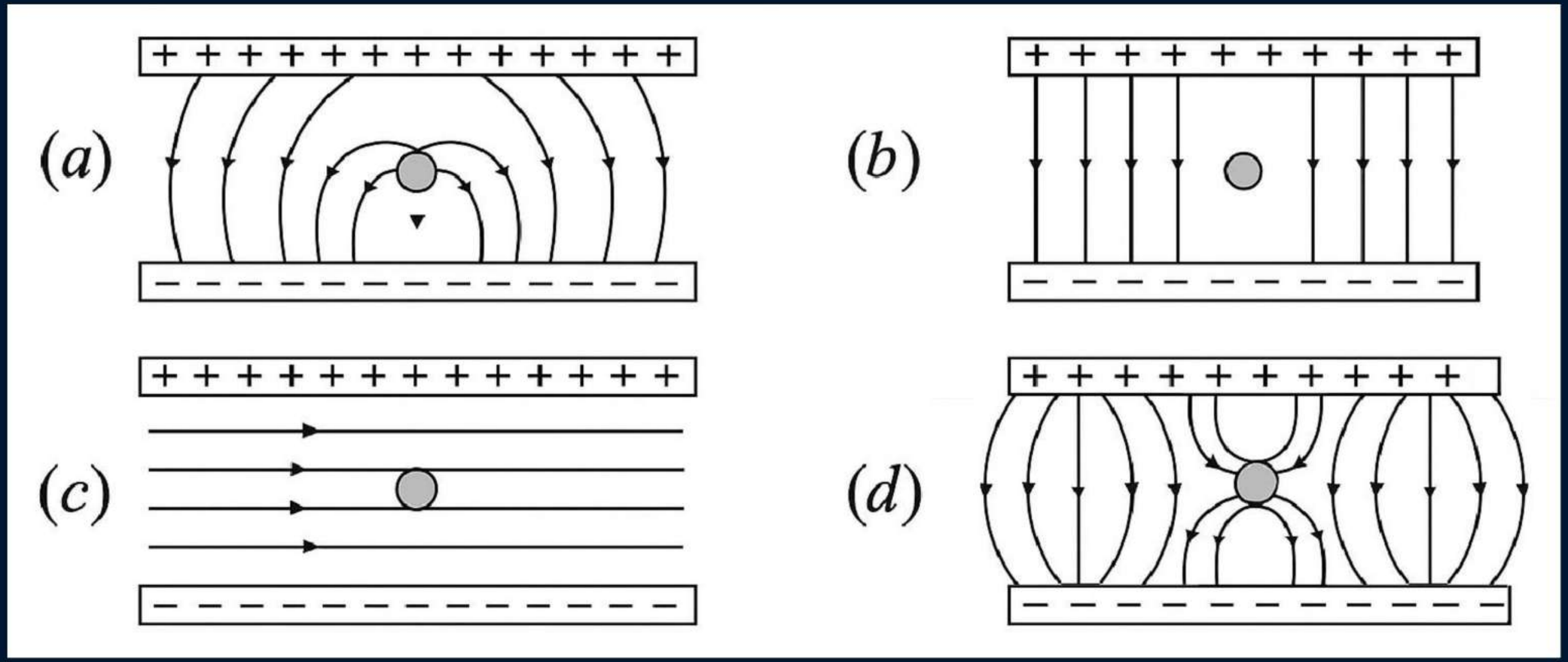
HW

Find the total charge stored in the network of capacitors connected between A and B as shown in figure: **(Ev) CBSE, 2020)**



# QUESTION

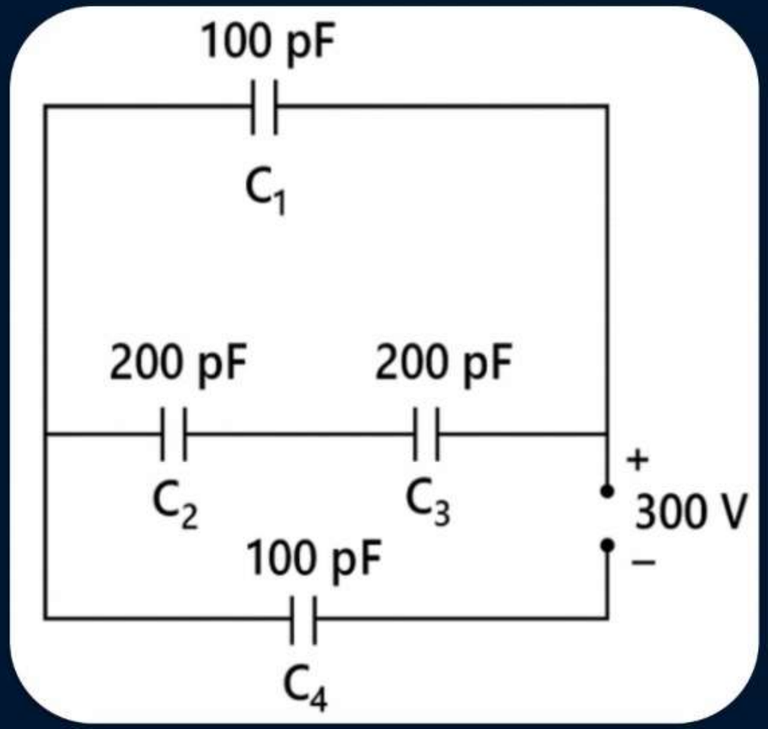
Which of the diagram correctly represents the electric field between two charged plates if a neutral conductor is placed in between the plates? **(An) (CBSE, 2022)**



# QUESTION

- Derive an expression for the capacitance of a parallel plate capacitor with air present between the two plates. **(2m)**
- Obtain the equivalent capacitance of the network shown in figure. For a 300 V supply, determine the charge on each capacitor. **(3m)**

**(An) (CBSE SQP, 2023)**






# QUESTION

Three capacitors $C_1$, $C_2$ and $C_3$ are connected in a combination as shown below. Identify the correct statement(s). (Un)

(i) The charge on capacitor $C_1$ is greater than that on capacitor $C_2$.

(ii) The charge on capacitor $C_1$ is the same as that on capacitor $C_3$.

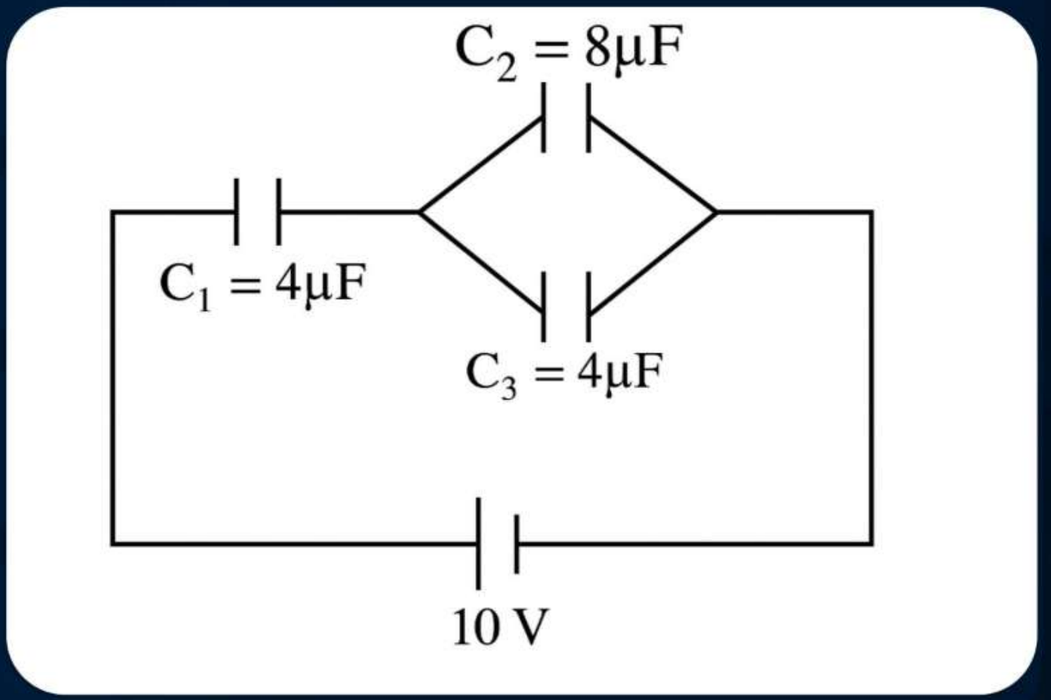
(iii) The charge on capacitor $C_1$ is 30 μC.

**A** Only (i) is correct.

**B** Only (iii) is correct.

**C** Both (i) and (iii) are correct.

**D** Both (i) and (ii) are correct.



Homework

↳ 6 Ques.

# THANK YOU

